

Contribution of monoamine oxidase (MAO) inhibition to tobacco and alcohol addiction.



Whole-body PET-scan studies in brains of tobacco smokers have shown a decrease in monoamine oxidase (MAO) activity, which reverts to control level when they quit smoking. The observed decrease in MAO activity in smokers is presumably due to their exposure to tobacco constituents that possess MAO-inhibiting properties. The inhibition of MAO activity seems, however, not to be a unique feature of tobacco smoking as subjects with Type II alcoholism have been reported to show a similar decrease in MAO activity that reverses when they cease to use alcohol. The present review summarizes the data on MAO-inhibiting tobacco constituents and explains that the decrease in MAO activity observed in alcoholics is probably due to concomitant tobacco use. It is concluded that the inhibition of MAO by constituents contained in tobacco and tobacco smoke, enhances the addiction induced by tobacco smoking.